

JOSEPH HAAS

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Senior Software Engineer | Technical Product Manager

Backend & API Development ▪ Payments & Benefits Systems ▪ Product Strategy & Delivery

Senior software engineer with five years of building and owning backend payments and benefits systems at Guild, following eight years as a product manager shipping software used by millions of customers. Works the full arc from problem discovery and API contract design through implementation, rollout, and incident response. Known as the engineer who writes the architecture doc, aligns the squads, ships the endpoint, and then traces the data when something goes wrong. Comfortable translating between product, operations, finance, and engineering, and most effective on systems where correctness and money are the same problem.

Areas of Expertise:

- Backend & API Design
- Distributed Systems
- System Architecture & ADRs
- Event-Driven Integration
- Performance & Load Testing
- Payments & Ledger Systems
- Data Migrations & Backfills
- Observability & Alerting
- Incident Response & Forensics
- Security Remediation
- Product Strategy & Roadmaps
- Requirements & User Stories
- Cross-Functional Leadership
- Agile & SAFe Delivery
- Technical Documentation & Runbooks

Technical Skills: Node.js, TypeScript, Python, SQL, GraphQL (AWS AppSync), REST API design, PostgreSQL, Snowflake, AWS (Lambda, AppSync, RDS), GitHub Actions (CI/CD, load-test automation), Mocha/Chai/Sinon, Snyk, event-driven consumers, Salesforce and NetSuite integrations, Jira, Confluence

Professional Experience

GUILD (formerly Guild Education) – Denver, CO

2020 – Present

Education and career-mobility benefits platform serving Fortune 1000 employers; Transaction Services owns the systems that fund, track, and reconcile every learner benefit dollar.

Senior Software Engineer, Transaction Services

2024 – Present

- Lead engineer for Transaction Services' integration with XOB (externalized benefits determination), the company-wide BA 2.0 migration off a legacy eligibility system: co-authored the createSpend-with-XOBs API contract, shipped XOB-based tax classification and contextual funding-limit derivation across the Funding Service and FACT, and built soft-launch parity checks that gate the production rollout with real data.
- Designed and shipped a bulk getStudentSpend endpoint for large-cohort consumers, then refactored it to batch fetches across all learners in a single query; built GitHub Actions load-test tooling to set baselines and track improvement.
- Led the spike that defined Guild's strategy for automated external spend ingestion; authored the Fidelity data contract (five event types, full batch lifecycle) and the implementation stories for the first integration.
- Contributed core functionality to EPOCH, a new employer calendar harmonization service: bidirectional propagation of funding-period changes, gap-tolerant ruleset logic, and integration test infrastructure.
- Hardened reliability for the employer payments portal and Funding Service: built metric-based alerting dashboards and runbooks that cut time to detect spend-check and report failures, replaced manual deploy checks with scripted smoke tests, and remediated critical and high vulnerabilities.
- Led incident investigations and data forensics across Funding Service and Snowflake (ghost enrollments, funding-cap overages, missing employee records), turning each into a reusable runbook.

Software Engineer II, Transaction Services

2022 – 2024

- Architected and shipped createSpend, the core spend-writing API for four benefit types (tuition assistance, reimbursement, professional expenses, student loan repayment) with multi-context funding determination, idempotency, and payor-aware writes, retiring a fragmented legacy pattern.
- Owned the FACT (Funding Activity Context Tracker) service from inception, building the endpoints that carry coverage package, academic partner, and eligibility context into every spend decision and downstream funding-limit calculation.
- Built Learner Spend Reports end to end (backend list/create/view API plus the employer-facing frontend) so employer partners can self-serve reports without engineering; wrote and ran NetSuite and reimbursement backfills for 10+ partners, including Walmart, Disney, Hilton, and Kohl's.
- Created a single control plane for tax eligibility across the transaction layer and replaced hardcoded 1099 tax-section rules with config-driven derivation, fixing misclassification for affected employers.
- Stood up the Transaction Registry foundation (auth scopes, read replicas, bulk update endpoint, payment-status event consumer), dual-wrote adjustment and UUID-merge events for a ledger-accurate view of spend, and authored the backfill and communication strategy for the migration.

- Integrated Funding Service with ePro for program enrollment validation, removing a cross-team dependency; evaluated and standardized the team's unit-test tooling on Mocha/Chai/Sinon.
- Authored the team's ADRs and architecture docs (member status tracking and cap determination, transaction locking, Transaction Registry strategy) and the incident runbook for hijacked accounts.

Software Engineer I, Tuition Assistance and Transaction Services

2021 – 2022

- Shipped eligibility correctness fixes in the E9Y distributed eligibility system (course lock behavior, as-of date propagation, course visibility) governing benefit decisions for thousands of students.
- Delivered the Salesforce cross-year reconciliation integration, enabling accurate end-of-year benefit accounting and preventing employer over- and underpayment across fiscal-year boundaries.
- Extended the Funding Service API (filtering, pagination, bulk adjustments, tax-period endpoints), integrated it into Guild's GraphQL gateway, and built the Manual Adjustments Tool so Operations could correct spend data without engineering.
- Onboarded nine employer partners (Humana, Tyson, Lowe's, Kohl's, Regions Bank, Pitney Bowes, and others) through benefit configuration and SQL data migrations, and wrote the runbooks that shortened every launch after; migrated security scanning from ShiftLeft to Snyk across five repositories.

Product Manager, Payment Products

2020 – 2021

- Owned balance tracking and external benefits administration products on the cross-functional payments scrum teams, serving Fortune 1000 employer partners and the systems that determine how each learner's benefit dollars are funded and reconciled.
- Wrote detailed user stories and front-end mockups, defined MVPs, and prioritized the backlog for engineering teams delivering payments features on a two-week cadence.
- Coordinated multi-team projects with fellow product managers and engineering leaders, aligning requirements, deployment sequencing, and cross-squad dependencies for releases spanning several services.

MJ FREEWAY – Denver, CO

2018 – 2019

Software Product Manager, Core Services

- Product owner for the Core Services suite (Cultivation, Processing, Point of Sale); defined the end-to-end enterprise development strategy on the SAFe framework and kept releases shipping every two to three weeks.
- Wrote requirements, user stories, wireframes, and UI mockups; collaborated with enterprise clients to validate roadmap milestones and reduce refactors.
- Trained the product organization on SDLC practices, from bug and story writing to identifying core customer needs; led the cross-functional evaluation of new global markets.

CHARTER COMMUNICATIONS – Denver, CO

2014 – 2018

Manager, Video Products

- Defined the roadmap and owned the backlog for three TiVo set-top video guide products supporting 20% of enterprise video customers; managed cross-platform features across set-top box, Roku, Xbox, Samsung, web, and mobile.
- Feature owner for accessibility compliance: partnered with UX to ship a text-to-speech guide usable without vision, plus a pre-installed Samsung tablet app and a web portal with cloud DVR.
- Won the Charter Champion Award (2015) for rebuilding internal and customer-facing support tools with the Care team, reducing call handling time and adding self-service options.

iBAHN – Denver, CO

2012 – 2014

Associate Product Manager, IPTV Platform / Hospitality Public PC Solutions

- Managed the ETVI in-room video platform for hotels (custom set-top hardware, live TV and on-demand UI, room-control integrations) and led marketing, business development, and global product development for a public-PC product for hotel lobbies and business centers.
- Gathered UI, custom-skin, and feature requirements for large clients including Marriott, Hilton, and Four Seasons; turned customer issues into feature requirements and better support and error handling.
- Analyzed market trends and competitor data to identify product gaps; directed an internship team in competitive research against the ETVI product.

Education & Certification

Back-End Engineering Program | Turing School of Software and Design, Remote | 2021

Bachelor of Science in International Business, minor in Marketing | University of Denver, Denver, CO | 2012

Certifications: Certified Scrum Product Owner ▪ Pragmatic Marketing Certification